Title: **Windowed Black Box Model and Multiple Channels Test Method Applied in Routing Protocol Test**

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Abstract: The protocol implementation was tested as black box before. Due to the complexity of routing protocols, the black box model can not fulfil the requirements to test them. Based on the analysis and perception of routing protocols, the windowed black box model is raised. The controllability and the observability are enhanced through other modules that interact with the module under test using standard interfaces. The multiple channels test method is put forward accordingly with more channels such as configuration channel and auxiliary information channel. The coverage and the efficiency of the test are increased considerably. With techniques like packet extension, the test suite is organized and described uniformly and integrally, which guarantees the flexibility and the universality and paves the way for the transition from single centered test suite to a distributed one. This model and this test method have achieved desirable results from routing protocol test and also give valuable references for the test of other protocols.

keywords: Reliability and fault-tolerance, Communication and telecommunication, Tools and environments for software development

1.Introduction

The implementation under test (IUT) of communication protocol is tested as black box to check its functions [1]. Only the output of the IUT responding to specific input is cared by the tester and no implementation details are necessary. This model leaves the programmers enough space and enables a general test system. With the rapid development of network techniques, network equipment (protocol implementation) possesses more abundant functions with a more complicated structure, demanding more power of protocol test to ensure its quality. The cost to perform the test will be increased considerably while maybe with poor efficiency and effect if the traditional black box model is still followed due to this complexity. On the other hand, the system containing the module under test (MUT) does offer the tester more information and more ability of control and observation. Therefore, a break through to the black box model is necessary and feasible.

In the test of the routing protocol, the state machine and packet structure need to be tested and can be performed with traditional test methods. Besides, the information process of routing protocol is also needed to be tested, which is much more important and maybe more difficult for testing routing protocol. If the IUT (or the MUT) is still treated just as a black box and is tested with traditional test method, test practice shows that the test case will be difficult to write and the efficiency will be quite low. Quite a lot of other protocol modules have been implemented in the router, such as SNMP module, telnet module and etc. Through the interface of SNMP, the routing table and the routing information base (RIB) of the routing protocol can be acquired and even configured [2]. While the router and the routing protocol can be configured through telnet [3]. With these standard interfaces of such modules, the observability and the controllability can be enforced in the routing protocol test, which will promote test efficiency and effect.

With consideration of these points and the experience from test practice, the implementation of routing protocol is tested as a windowed black box. That is, besides the input and output of the black box, the tester can observe and control it through other modules that interact with it (each of them is called a window). To test a

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windowed black box, the tester still need not know the details of implementation. Accordingly, the multiple channels test method is brought out. Not only does the test system interact with the module under test (MUT) of the protocol (this is called a basic channel), it also controls and observes the MUT through other channels (these are the extended channels). Thus the efficiency and the effect of test are upgraded. Since the extended channels still take advantage of the standard interfaces, the multiple channels test method is still a generic one.

With the windowed black box and the multiple channels test method, the test of BGP-4 [4], OSPF [5] and RIP [6] is carried out and valuable results are acquired. This method is also applied to test quite a lot of other protocols and plays an important role in router test. In this paper we will mainly illustrate the test of BGP-4 as a case study.

This paper consists of five sections. In section 2, the features and the test requirements of routing protocols are discussed. In section 3, the windowed black box and the multiple channels test method are presented. In section 4, the test practice is explained. Finally, related work and conclusion are given.

2.Features and Test Requirements of Routing Protocols

The clear perception of routing protocols features is the basis of the test activity. Compared with communication protocols such as TCP/IP, routing protocols are quite different in several aspects. First, the purpose of routing protocol is to collect the information needed to do routing [6], which is a basic summery of routing protocols functions. Another extraordinary character of routing protocols is the nature of distributed systems, which can be found from both the interaction between different routers and the communication between the interfaces of one single router. No matter from the design or from the implementation, there are several important databases to store the information. The information flow among these databases is connected by the internal processes of routing protocols. These processes dominate the correctness and efficiency of routing protocols implementations. Besides, the "level" of routing protocols is quite different from that of communication protocols: for a N level communication protocol, it always takes advantage of the service of N-1 level protocol and offers service for N+1 level. While for a routing protocol, it always exploits the lower level to exchange routing information and there is no "upper level protocol" at all. However the interaction of routing protocols is much more complex which is composed of two important types. For example, BGP-4 will talk with the neighbor peers while exchange information with other routing protocol such as OSPF in the same router. The detailed discussion of routing protocols features can be referred to [7].

For the test of communication protocols, the protocol state machine is heavily relied on from test suite design and generation to the coverage analysis. Although there are state machines of routing protocols, they only describe a little part of routing protocols behavior. The majority of routing protocols activities begins after they enter a "stable" state (say, Established state for BGP-4). The reason is that routing protocols will process the information exchanged instead of just transmit it.

Therefore, the test of routing protocol should cover (1) packet structure, (2) protocol state machine, (3) interoperability, (4) exchange and manipulation of routing information and (5) performance analysis and evaluation.

Four abstract test methods are recommended in [1] for end system test, which are Local Test Method (LTM), Distributed Test Method (DTM), Coordinated Test Method (CTM) and Remote Test Method (RTM). These methods are mainly used to test communication protocols. If this classification is followed, then only the test method like RTM can be used to test routing protocol [7].

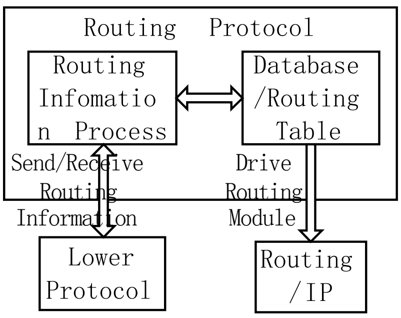

Figure 1. Architecture of the routing protocol

In the test of communication protocol, the MUT is tested as a black box. However, for routing protocol, its purpose is to collect the information needed to do routing (as shown in figure 1), receiving and sending routing information with lower level protocol. All these information will be processed by the routing information process module and stored in corresponding databases. The routing table, which may also be regarded as a database, drives the IP level to do the packet forwarding. If we follow the old method and use only one channel, what we will do will be to first observe the behavior of IP forwarding, then deduce the correctness of routing table and finally conclude the correctness of routing information exchange and manipulation. This will inevitably lead to a complex test suite and limit the test ability. Therefore, the black box model is no longer suitable to test the routing information process of routing protocol.

The interoperability of the routing protocol, such as the interaction between BGP-4 peers, can be tested online as in [8]. Test of the performance will not be discussed in this paper.

Current test method, test system and achievements shall be taken good use of to do what they can do in routing protocol test. PITS [9] has achieved a lot in communication protocol test. Therefore it can undertake the test of packet structure and protocol state machine. While to the test of routing information manipulation, new models, methods and techniques are needed as discussed above. And necessary extension or improvement should be made to current test system, such as PITS. All these will be discussed in the following sections.

3. Windowed Black Box Model and Multiple Channels Test Method

3.1 Windowed Black Box Model

Black box model is the most widely exploited one in protocol test to describe the MUT. The main idea of black box is that the tester only need care about the input and output of the MUT while needn't know the implementation details, which is suitable for function test. Protocol test is a branch of software test in which two more models white box model and gray box model are also used. The general description will be something like this [10]:

White box model: the tester is able to extract test data from the analysis of program code and the information of the code and the components of the program. It is a structure test.

Black box model: Test data is extracted from program description. It is a function test.

Gray box model: That between the above two is a gray box test.

Now a more formal definition will be given:

A module under test (MUT) is a 6-tuple: **<I,O,D,P(I),P(O),P(D)>**

Where **I** and **O** are the sets of input data and output data of the MUT. **D** is the set of internal data. **P(I)**, **P(O)** and **P(D)** are the sets of the processes dealing with input data set **I**, output data set **O** and internal data set **D**. Then

(1) If the tester can (or have to) know all these six elements, then the tester tests the MUT as a white box.

(2) If the tester can (or need) only know the input data set **I** and output data set **O**, then the tester tests the MUT as a black box.

(3) If the tester can (or have to) know the elements between (1) and (2), then the tester tests the MUT as a gray box.

For example, the corresponding six elements of routing protocol are routing information received, routing information sent out, routing information databases (including routing table, database and internal parameters), routing information receiving processes, routing information sending processes and routing information manipulation processes (including the route preference calculation, selection, database update and etc).

All these above definitions and classifications only focus on the MUT itself. While the windowed black box takes into consideration not only the MUT but also other modules interacting with the MUT to control and observe the data. As an example, besides the routing protocol, one router has other modules implemented, such as SNMP module and telnet module. These modules interact with the routing protocol module and can access the routing information database and configure the router. These modules offer standard interfaces and provide the tester several windows to enhance his observation and control of the MUT of routing protocol. Therefore, the windowed black box is defined to support the routing protocol test:

If tester can (or have to) know the input data set **I,** output data set **O** and window set **W**, then the tester tests the MUT as a windowed black box. Where $\mathbf{W = \{W_1, W_2, \ldots , W_k\}}$ are the windows provided by other modules $\mathbf{M_1}$, $\mathbf{M_2}$, …, $\mathbf{M_k}$ that interact with the MUT.

The windowed black box still possesses the property of the black box, that is, only the description of the MUT functions is cared while the implementation details are not. The access through windows also takes advantage of the standard interfaces, no details of implementation are necessary either. If W is empty, then the windowed black box retrogresses to be a black box.

3.2 The Multiple Channels Test Method

Accommodating with the windowed black box, the multiple channels test method is exploited in the test of routing protocol. In traditional work, the test system only setups communication channel with the MUT (although there may be more than one channels, all these channels are restricted to the MUT only), we call this channel a "basic channel". Through basic channel, the test system can inject input and observe its output. The typical example can be found in [11]. The principal difference of the multiple channels test method is that not only does the test system setup communication channel with the MUT, it also setups channels with other modules interacting with the MUT (as shown in figure 2).

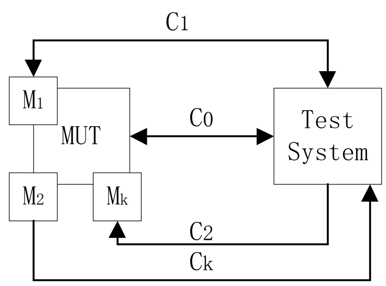

Figure 2. Multiple channels test method and test architecture

The $\mathbf{C_0}$ is the basic channel between the test system and the MUT. $\mathbf{M_1, M_2, \ldots, M_k}$ are the k modules that interact with the MUT. The test system setups channels $\mathbf{C_1}$, $\mathbf{C_2}$, …, $\mathbf{C_k}$ with these modules. Without these channels, the multiple channels test method goes back to be the original one.

The main idea of the multiple channels test method is to take full advantage of the information available to enhance the observability and the controllability so that test efficiency and test effect can be promoted. However, for a long time, protocol test only pays attention to the single MUT while this information is not exploited at all.

The multiple channels test method requires some precondition to take effect: the correctness of other modules. This can be achieved by testing $\mathbf{M_1, M_2, \cdots, M_k}$ in advance. Actually, testing those functions that will be used is enough.

4 Practice of Routing Protocol Test

The windowed black box and the multiple channels test method are applied in routing protocol test. The implementation of routing protocol always resides in the router. Therefore other modules such as SNMP and telnet are also implemented. Network management software can access information through SNMP, such as the number of interfaces, status, uptime, … and especially the information that are interested in routing protocol test. On the other hand, telnet provides means for user to check and configure the router remotely. As to routing protocol test, it can be used to modify the configurations such as AS number, router ID, routing policy and etc. Then we can test the correctness of routing protocol in different conditions.

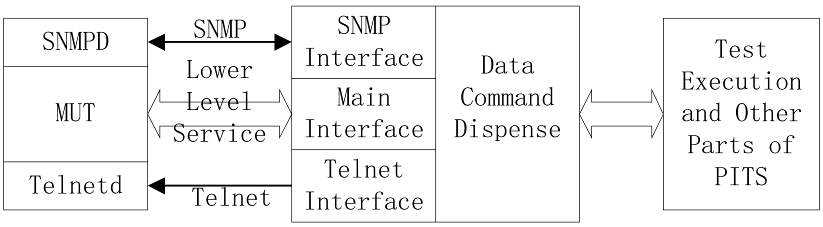

Figure 3. The test architecture applied in routing protocol test

Based on this, the whole test architecture is extended. Besides the basic channel between the test system and the MUT, configuration channel (telnet) and auxiliary information channel (SNMP) are introduced (shown in figure 3). The auxiliary information channel uses service provided by SNMP to get and set the interested information and the configuration channel changes the router configuration with corresponding commands through telnet. A configuration script is illustrated in figure 4, which will be sent to the router through configuration channel to change the BGP-4 peers from external peers to internal peers.

The lower level service used by the basic channel depends on the protocol under test: exploiting TCP [4] when testing BGP-4, IP and linker level service for OSPF[6] and UDP for RIP.

These different data used by three channels is dispensed to corresponding modules so that they can cooperate to perform the task. These modules function together as the reference implementation (RI) of the original test system. Due to space limitation, the test execution and other parts of the test system are not depicted in figure 4.

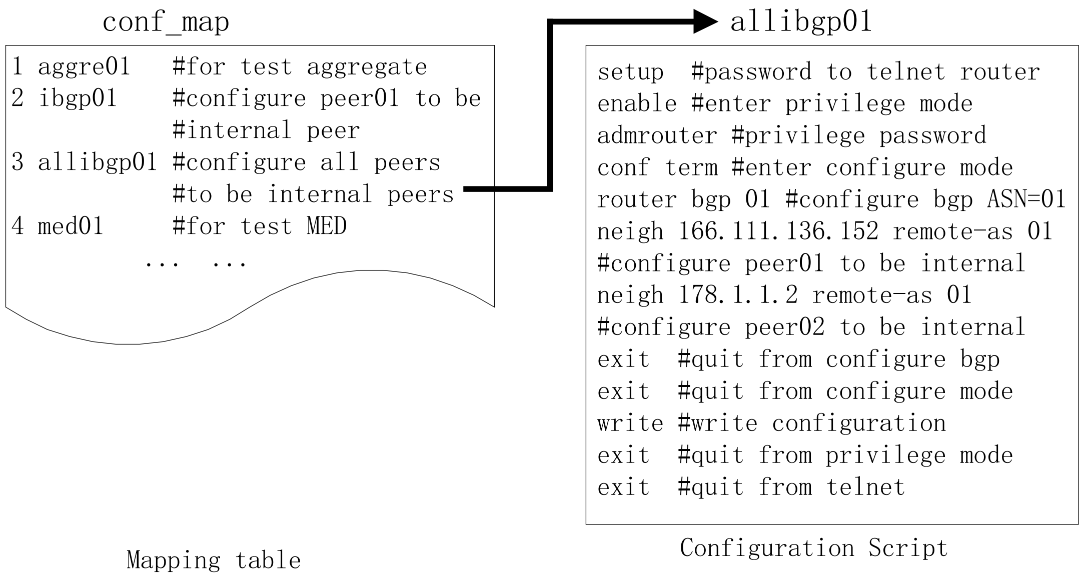

Figure 4. The mapping table and test configuration script for the configuration channel

The detail can be referred to [9]. Therefore, the functions of test system are enhanced while its compatibility and stability are maintained.

The multiple channels test method endues the test system with more ability but also cause new problems to solve. One of the problems is how to describe the test suite uniformly and integrally: the total amount increases a lot with all the data of these three channels. While these data is different in purpose and format but needs cooperation to perform the test. Besides, the test suite is always described in TTCN (Tree and Tabular Combined Notation) and it is quite difficult to use TTCN to represent the information used by the configuration channel and the auxiliary information channel. Therefore, the packet expansion technique is raised to overcome this. For example, four types of BGP-4 packets are defined in [4] and we defined type 128-143 for the date sent through the configuration channel and 144-161 for auxiliary information channel. This definition does not contradict with current usage of packet type and is unlike to cause confusion for a quite long time in the future. However, there is no actual data in the date filed of the packet with special type value. There is only an index and the data will be looked up by corresponding interfaces in their mapping tables (shown in figure 4). This form of representation is a natural extension from the PIXIT (Protocol Implementation eXtra Information), which can act as a external part of the test suite. Thus the test suite is described integrally and uniformly and paves way for the transition from single centered test suite to a distributed one. An example of a test case is shown in figure 5, in which the configuration channel and the auxiliary channel are exploited.

Dynamic Part					
Test Case Name:	UPDATE_ALL_IBGP				
Group:	bgp-4/IBGP/UPDATE_ALL_IBGP				
Purpose:	Test the information manipulation of IBGP				
Default:	NULL				
Comments:	NULL				
No	Behavior Description	Label	Constraints Ref	Verdict	Comments
1	T! TO_CONF(TO_CONF.NO :='0003'O)		C_TO_CONF		Config through conf channel
2	+ ESTABLISHED				Test step to Established state
3	T! UPDATE01		C_UPDATE01		Send through main channel
4	T! UPDATE01		C2_UPDATE01		Send another
5	START T_HOLD				Start Hold Timer
6	T? FROM_AUX		C_FROM_AUX		Get infor from aux channel
7	START T_KEEP				Start Keepalive Timer
8	T? KEEP_ALIVE01		C_KEEPALIVE01	PASS	Keepalive arrives in time
9	? TIMEOUT			FAIL	Nothing received in time
10	T? OTHERWISE			FAIL	Other packet received instead
11	? TIMEOUT			FAIL	Nothing received in time
12	T? OTHERWISE			FAIL	Other packet received instead
Extended Comments: NULL					

Figure 5. An example of a test case

With only the basic channel, the test mainly can deal with the packet structure and protocol state machine. With the auxiliary information channel added, the test coverage is increased. These two conditions are compared in table 1 with the test of BGP-4 as an example.

Table 1 Comparison of test content covered (number in the bracket are the amount of test cases)

Test Content / Channel	Routing Information Base	Receiving Process	Sending Process	Calcualtion and Selection	Convergence Property	Kernel Routing Table
Only with basic channel	Unable to test	Unable to test	Unable to test	Limited to those sent out (35)	Unable to test	Limited to those sent out (27)
Basic Channel + auxiliary information channel	Able to test (52)	Able to test （34）	Able to test (23)	No matter sent out or not（82）	Able to test （17）	No matter sent out or not（58）

With auxiliary information channel, those aspects unable to test before are testable now and more test can be done to those already testable originally which is reflected in the increase of the number of test cases. As to the test of packet structure and protocol state machine, since it doesn't need other channels, the test coverage and the number of test cases remain unchanged. Therefore, we can say the ability of routing protocol test is strengthened.

On the other hand, the effect of the configuration channel mainly reflects in the promotion of test efficiency. Still with BGP-4 test as an example, in the tests of the same MUT (of a router), all the 362 test cases are executed without and with the configuration channel separately. The time expended is compared in table 2.

Table 2. Time comparison of the test

Test Content / Channel	**Packet Structure**	**Protocol State Machine**	**Routing Information Manipulation**	**Total**
Basic Channel + Assitant Information Channel	About 60 min.	About 30 min.	About 770 min.	About 860 min.
Basic Channel + Assitant Information Channel + Configuration Channel	About 60 min.	About 30 min.	About 360 min.	About 450 min.

As to the tests of packet structure and protocol state machine which doesn't need other channels, the time expended is not increased. To those exploiting other channels, the time is reduced to below 50% of the original time and the total time is decreased to about 50%. The main factor of time expense cut down is the time to configure the router by hand is saved, which also increases the stability and prevents human errors.

Another effect of the configuration channel is that the test execution can be carried out automatically without human interference or assistance, which is a prerequisite for the automation of the whole test activity and paves the way for large scale and continuous router test.

Before we test the routing protocol in the router, we have already had SNMP and telnet tested as in [12] to assure their correctness.

During the test practice, it is revealed that we can expect more from those "accomplished" protocol tests and those encountering difficulties before. Now the efficiency and the effect are increased with multiple channels test method. For example, the IP test [11] only covered a limited scope of IP forwarding with test method of that time, now we can test it in more details. Till the test of TCP in the router, there are some difficulties encountered then

[13], now the new test method overcomes them. These will be discussed elsewhere due to space limitation.

5 Related Work and Conclusions

Currently, routing protocol test is a hot spot of protocol test. Due to the difference between routing protocols and communication protocols, no matter test method or test system suitable to test communication protocols before is confronted with challenge. Researchers have done quite some work in this field. Reference [14] discussed the test of OSPF using Concurrent TTCN (C-TTCN) and Concurrent EBE (C-EBE), but it mostly focused on the test of the *state machine* almost without anything involved with the *routing* aspect of the protocol. Reference [15] presents the test of IP routing protocols with the test environment generated by the software tool SOCRATES. Reference [16] is the previous work we carried out on PITS with only the basic channel. Reference [8] describes the intelligent online BGP-4 analyzer we designed and implemented.

In the test practice, because routing protocols possess complex information processes, the original black box model and test method exploited encounter difficulties. The reason is that the old method only sets up communication channel with the MUT only. Therefore, the test ability is completely restricted to those can be observed/controlled on this channel. With the development of network techniques, network equipment is getting more mature with more functions and provides more ability to control and observe. However, this available information and the power haven't been used for a long time. Therefore, this paper starts from the black box model and raises the windowed black box model. Its main idea is to enhance the ability to control and observe through other modules interact with the MUT. Based on this, the multiple channels test method is put into use. Besides the channel between the test system and the MUT, other channels exploiting standard interfaces are added, such as auxiliary information channel with SNMP and configuration channel with telnet. In order to describe and organize the test suite with data of several channels, the packet extension technique is put forward to represent all the data in the test suite uniformly and integrally with compatibility. Besides, the multiple channels test method promotes automation of test activity, that is, no human interference or assistance is necessary during test execution.

With these techniques, the test scope is enlarged and the time expense is lowered considerably. Therefore, the test efficiency and test ability are improved.

All these discussed have been applied in routing protocol test. They are also useful for those tests "accomplished" or those with difficulties before. The windowed black box and the multiple channels test method also pave way for the transition from single module test to system test, especially for the imperative router test that is also the focus of our future work.

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